Section 3.3- Expected value and Variance

Intuition: Given a random variable, the expected value tells us the center of mass of a distribution. The expected value is the weighted average of the values that random variable can take, weighted by their probabilities. Most of the time in statistic when they say average, they mean the expected value.

Definition (Expected Value=mean=average)

Let X be a discrete random variable that takes values $x_1, x_2, x_3, \cdots$. (finite or countably infinite).
Let $p_X(x)$ be the pmf of X, then the expected value of X, denoted by $E[X]$ or μ is defined as

$$E[X] = \sum_{i=1}^{n} x_i \cdot p_X(x_i)$$

Note: Expected value is always a number.

Properties of Expectation: Expectation is always a linear function meaning:

(a) $E[X+Y] = E[X] + E[Y]$, for any random variables X and Y.

(b) $E[aX] = a \cdot E[X]$

(c) $E[c] = c$

(d)

$$E[f(X)] = \sum_{i=1}^{n} f(x_i) \cdot p_X(x_i)$$

Eg 1. You test electrical components to determine whether they work or not.

$$X = \begin{cases} 1 & \text{if the component works(success)} \\ 0 & \text{if the component fails} \end{cases}$$

Define the probability mass function with

$$p(x) = \begin{cases} 0.1 & \text{if} x = 0 \\ 0.9 & \text{if} x = 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the expected value of X.

$$E[X] = \sum_x x \, P_X(x)$$

$$E[X] = 0 \cdot P(X=0) + 1 \cdot P(X=1) = 0 + 1(0.9)$$

$$= 0.9$$

Eg 2-1. Suppose a random variable X has the following probability distribution:

x	$p_X(x)$
1	0.2
2	0.4
3	0.3
4	0.1

(a) $E[X]$.
(b) $E[X^2]$.
(c) $E[3X + 4]$.

$$E[X] = \sum_x x\, p_X(x) = 1 \cdot p_X(x=1) + 2 \cdot p_X(x=2)$$

$$+ 3 \cdot p_X(x=3) + 4 \cdot p_X(x=4)$$

$$= 1 \cdot (0.2) + 2 \cdot (0.4) + 3 \cdot (0.3) + 4 \cdot (0.1) =$$

$$= 0.2 + 0.8 + 0.9 + 0.4 = 2.3$$

$$\boxed{E[X^2]} = \sum_x x^2 \, P_X(x)$$

$$= (1)^2 \cdot 0.2 + (2)^2 \cdot 0.4 + (3)^2 \cdot 0.3 + 4^2 \cdot (0.1)$$

$$= \overbrace{0.2 + 1.6 + 2.7}^{2.9} + 1.6 = \boxed{6.1}$$

$$= 0.2 + \underbrace{1.6 + 2.7 + 1.6}_{3.2}$$

$$\boxed{E[3X+4]} = 3\,E[X] + 4 = 3 \cdot (2.3) + 4$$

$$= 6.9 + 4 = \boxed{10.9}$$

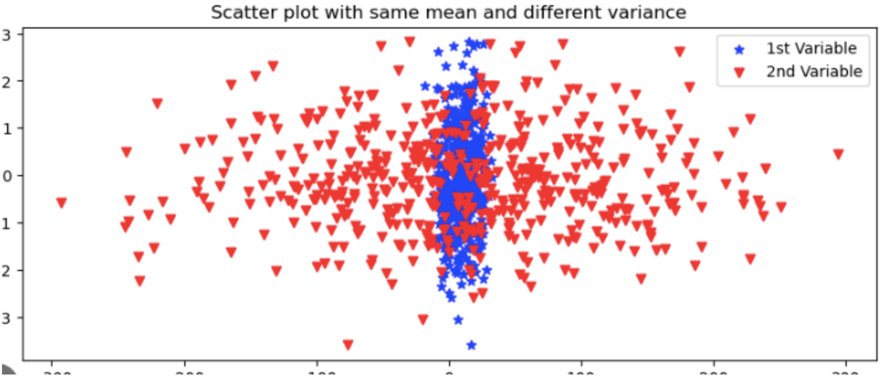

Figure: Variance shows how spread out our data is.

Intuition: Given a random variable, the variance tells us how spread out the distribution is.
The variance of X measures how far X is from its mean μ on average, but instead of simply taking the average difference between X and its mean μ, we take the average squared difference.

Definition (Variance)

The variance of an r.v. X is

$$\sigma^2 = Var(X) = E[(X - E[X])^2] = E[(X - \mu)^2]$$

Var(X) can also be calculated from the below

$$\sigma^2 = Var(X) = E[X^2] - E[X]^2$$

The square root of the variance is called the standard deviation (SD):

$$\sigma = SD(X) = \sqrt{Var(X)}$$

(1) $Var(aX) = a^2 \cdot Var(X)$

(2) $Var(aX + b) = a^2 \cdot Var(X)$

(3) $SD(aX) = |a| SD(X)$

(4) $Var(b) = 0$

Eg 2-2. Suppose a random variable X has the following probability distribution:

x	$p_X(x)$
1	0.2
2	0.4
3	0.3
4	0.1

Find Var(X), Var(2X), SD(4 X), CDF(X).

$$\text{var}(x) = E[x^2] - (E[x])^2$$

we calculated E[X] and E[x²] on page 5 and 6.

$$E[X] = 2.3 \qquad E[x^2] = 6.1$$

$$\text{var}(x) = 6.1 - (2.3)^2 = 6.1 - 5.29 = 0.81$$

$$\text{var}(2x) = 4 \cdot \text{var}(x) = 4 \cdot (0.81) = 3.24$$

$$SD(4X) = 4\,SD(X) = 4 \cdot \sqrt{0.81} = 4 \cdot 0.9 = 3.6$$

For the cdf of X, let's show it with $F_X(x)$

$$F_X(x) = P(X \leq x) = \begin{cases} 0 & x < 1 \\ P_X(1) = 0.2 & 1 \leq x < 2 \\ 0.2 + P_X(2) = 0.2 + 0.4 = 0.6 & 2 \leq x < 3 \\ 0.6 + P_X(3) = 0.6 + 0.3 = 0.9 & 3 \leq x < 4 \\ 0.9 + P_X(4) = 0.9 + 0.1 = 1 & 4 \leq x \end{cases}$$

$$F_X(x) = \begin{cases} 0 & x < 1 \\ 0.2 & 1 \leq x < 2 \\ 0.6 & 2 \leq x < 3 \\ 0.9 & 3 \leq x < 4 \\ 1 & 4 \leq x \end{cases}$$

Eg 3. Suppose we roll a pair of fair 6-sided dice. Let X denote the maximum (the largest) of the outcomes on the two dice.
Find pmf of X and compute its expected value.

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

X can take values 1, 2, 3, 4, 5, 6.

x	1	2	3	4	5	6
$p_X(x)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

$$E[X] = \sum_x x \cdot p_X(x)$$

$$E[X] = 1 \cdot \frac{1}{36} + 2 \cdot \frac{3}{36} + 3 \cdot \frac{5}{36} + 4 \cdot \frac{7}{36} + 5 \cdot \frac{9}{36} + 6 \cdot \frac{11}{36}$$

$$E[X] = \frac{1 + 6 + 15 + 28 + 45 + 66}{36} = \frac{161}{36} = 4.472$$

Eg 4. Consider $p_X(x) = a(1+x)^2$ for $x = 0, 1, 2, 3$.
(a) Find a such that $p_X(x)$ is a pmf for the random variable X.
(b) E[X]
(c) SD(X)

(a) $\sum_x p_X(x) = 1 \Rightarrow p_X(0) + p_X(1) + p_X(2) + p_X(3) = 1$

$a + 4a + 9a + 16a = 1$

$30a = 1 \Rightarrow a = \frac{1}{30}$

(b) $E[X] = \sum_x x \cdot p_X(x) = 0 \cdot p_X(0) + 1 \cdot p_X(1) + 2 \cdot p_X(2)$
$+ 3 p_X(3) =$

$$E[X] = 1 \cdot \frac{4}{30} + 2 \cdot \frac{9}{30} + 3 \cdot \frac{16}{30} = \frac{4+18+48}{30} = \frac{70}{30} = \frac{7}{3} = 2.33$$

$$E[X^2] = 1 \cdot \frac{4}{30} + 4 \; \frac{9}{30} + 9 \cdot \frac{16}{30} = \frac{4+36+144}{30} = \frac{184}{30} = \frac{92}{15} \sim 6.133$$

$$\begin{aligned} Var(X) &= E[X^2] - (E[X])^2 \\ &= \frac{92}{15} - \left(\frac{7}{3}\right)^2 = \frac{92}{15} - \frac{49}{9} = \frac{276-245}{45} = \frac{31}{45} \end{aligned}$$

$$SD(X) = \sqrt{\frac{31}{45}} = 0.83$$